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Batch - 15

Experiment - 14

Question

Write a **menu-driven C program using switch-case** to input a positive integer and perform one of the following operations based on the user's choice:

1. Check whether **all digits are even**.
2. Check whether **all digits are prime**.

1. Algorithm

Start

1. Declare integer variables `num`, `choice`, `temp`, `digit` and `flag`.
2. Display the menu:
 - o 1. Check whether all digits are even
 - o 2. Check whether all digits are prime
3. Input the user's choice.
4. Input a positive integer `num`.
5. Use `switch(choice)`.

Case 1: Check whether all digits are even

6. Set `temp = num` and `flag = 1`.
7. Repeat while `temp > 0`:
 - o Find the last digit using `digit = temp % 10`.
 - o Check `digit % 2 != 0`.
 - o If true, set `flag = 0` and break the loop.
 - o Remove the last digit using `temp = temp / 10`.
8. If `flag == 1`, display "**All digits are even**".
9. Otherwise, display "**All digits are not even**".

Case 2: Check whether all digits are prime

10. Set `temp = num` and `flag = 1`.
11. Repeat while `temp > 0`:
 - o Find the last digit using `digit = temp % 10`.

- Check whether the digit is 2, 3, 5, or 7.
 - If the digit is not one of these, set `flag = 0` and break the loop.
 - Remove the last digit using `temp = temp / 10`.
12. If `flag == 1`, display "**All digits are prime**".
13. Otherwise, display "**All digits are not prime**".

Default

14. Display "**Invalid choice**" if the user enters a choice other than 1 or 2.
15. Stop.

2. Test Case Table

Test Case	Inputs	Expected Outputs	Actual Outputs	Results
1	Choice = 1, Number = 24680	All digits are even	All digits are even	Pass
2	Choice = 1, Number = 12345	All digits are not even	All digits are not even	Pass
3	Choice = 1, Number = 8642	All digits are even	All digits are even	Pass
4	Choice = 2, Number = 2357	All digits are prime	All digits are prime	Pass
5	Choice = 2, Number = 2379	All digits are not prime	All digits are not prime	Pass
6		All digits are prime	All digits are prime	Pass

3.C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
    int num, choice, temp, digit, flag;
    printf("1. Check whether all digits are even\n");
    printf("2. Check whether all digits are prime\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    printf("Enter a positive integer: ");
    scanf("%d", &num);
    switch(choice)
    {
        case 1:
            temp = num;
```

```

    flag = 1;
    while(temp > 0)
    {
        digit = temp % 10;
        if(digit % 2 != 0)
        {
            flag = 0;
            break;
        }
        temp = temp / 10;
    }
    if(flag == 1)
        printf("All digits are even\n");
    else
        printf("All digits are not even\n");
    break;
case 2:
    temp = num;
    flag = 1;
    while(temp > 0)
    {
        digit = temp % 10;
        if(digit != 2 && digit != 3 &&
            digit != 5 && digit != 7)
        {
            flag = 0;
            break;
        }
        temp = temp / 10;
    }
    if(flag == 1)
        printf("All digits are prime\n");
    else
        printf("All digits are not prime\n");

    break;
default:
    printf("Invalid choice\n");
}
return 0;
}

```

4. Compilation and Execution

```
(kali㉿kali)-[~]
└─$ gcc hdoc15.c -o hdoc15

(kali㉿kali)-[~]
└─$ ./hdoc15
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 24680
Enter a positive integer: r
Invalid choice

(kali㉿kali)-[~]
└─$ gcc hdoc15.c -o hdoc15

(kali㉿kali)-[~]
└─$ ./hdoc15
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
Enter a positive integer: 24680
All digits are even
```

```
(kali㉿kali)-[~]
└─$ gcc hdoc15.c -o hdoc15

(kali㉿kali)-[~]
└─$ ./hdoc15
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
Enter a positive integer: 12345
All digits are not even

(kali㉿kali)-[~]
└─$ gcc hdoc15.c -o hdoc15

(kali㉿kali)-[~]
└─$ ./hdoc15
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
Enter a positive integer: 8642
All digits are even

(kali㉿kali)-[~]
└─$ gcc hdoc15.c -o hdoc15

(kali㉿kali)-[~]
└─$ ./hdoc15
1. Check whether all digits are even
2. Check whether all digits are prime
```

```
Enter your choice: 2
Enter a positive integer: 2357
All digits are prime

(kali㉿kali)-[~]
└─$ gcc hdoc15.c -o hdoc15

(kali㉿kali)-[~]
└─$ ./hdoc15
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
Enter a positive integer: 2379
All digits are not prime

(kali㉿kali)-[~]
└─$ gcc hdoc15.c -o hdoc15

(kali㉿kali)-[~]
└─$ ./hdoc15
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
Enter a positive integer: 7532
All digits are prime
```